PRACTICAL - 3

Implementation of Max-Heap Sort Algorithm

Aim of the Practical:

To implement Heap Sort using a max heap and analyze its running time step by step.

Algorithm:

1. Build a max heap from the input array.
2. Swap the maximum element at the root with the last element.
3. Reduce heap size by one.
4. Heapify the root.
5. Repeat until the heap becomes empty.

Program:

```
#include <iostream>
using namespace std;
void heapify(int a[], int n, int i) {
    int largest = i;
    int left = 2*i + 1;
    int right = 2*i + 2;
    if (left < n && a[left] > a[largest]) largest = left;
    if (right < n && a[right] > a[largest]) largest = right;
    if (largest != i) {
        swap(a[i], a[largest]);
        heapify(a, n, largest);
    }
}
void heapSort(int a[], int n) {
    for (int i = n/2 - 1; i >= 0; i--)
        heapify(a, n, i);
    for (int i = n - 1; i > 0; i--) {
        swap(a[0], a[i]);
        heapify(a, i, 0);
    }
}
int main() {
    int a[] = {12, 11, 13, 5, 6, 7};
    int n = sizeof(a) / sizeof(a[0]);
    heapSort(a, n);
    for (int x : a) cout << x << " ";
}
```

Output:

```
5 6 7 11 12 13
```

Time Analysis:

Derivation of Time Complexity:

Heap Sort has two main phases: build the max heap and repeatedly extract the maximum.

Phase 1 - Build Heap: heapify is called on internal nodes from n/2 - 1 down to 0.

The tighter analysis of build-heap sums the work by node height:

$$T_{build}(n) = \Sigma(h = 0 \text{ to } \log n) [\text{number of nodes of height } h] \cdot O(h)$$

$$T_{build}(n) = O(n \cdot \Sigma(h \geq 0)\ h / 2^{(h+1)})$$

The infinite series $\Sigma\ h/2^{(h+1)}$ converges to a constant, therefore:

$$T_{build}(n) = O(n)$$

Phase 2 - Repeated extraction: there are (n - 1) extractions.

Each heapify after extraction takes at most O(log n).

$$T_{extract}(n) = (n - 1) \cdot O(\log n) = O(n \log n)$$

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Total:

$$T(n) = O(n) + O(n \log n) = O(n \log n)$$

Final Time Analysis:

- Best Case: Θ(n log n).
- Average Case: Θ(n log n).
- Worst Case: Θ(n log n).
- Auxiliary Space: O(log n) with recursive heapify.

Result:

Max-Heap Sort was implemented successfully and its total complexity was obtained by analyzing build-heap and extraction separately.